

Bluestock Mutual Fund Analytics

Final Capstone Report

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GitHub: [vinodkumarvasantha/Mutual-Fund-Analytics](https://github.com/vinodkumarvasantha/Mutual-Fund-Analytics)

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1. Executive Summary

This report presents the findings of a comprehensive mutual fund analytics capstone project covering a universe of 40 Indian mutual fund schemes across Equity and Debt categories. The project combined data engineering, statistical performance analysis, and business intelligence dashboarding to evaluate fund performance, investor behavior, and industry trends over the period 2022–2026.

Using four years of daily NAV history, industry AUM data, SIP inflow trends, and investor transaction records, this analysis computed risk-adjusted performance metrics (Sharpe Ratio, Sortino Ratio), benchmark-relative measures (Alpha, Beta), and drawdown statistics for all 40 schemes. A composite 0–100 Fund Scorecard was developed to rank funds across five weighted dimensions: 3-year returns, Sharpe Ratio, Alpha, expense ratio, and maximum drawdown.

The top-performing funds by composite score were UTI Mid Cap Fund, HDFC Top 100 Fund, and Axis Small Cap Fund. Industry-level findings showed total AUM of approximately ₹62.74 lakh crore, monthly SIP inflows of ₹31,002 crore, and 26.12 crore investor folios as of the most recent reporting period.

A four-page interactive Power BI dashboard was built to make these findings accessible to non-technical stakeholders, covering industry overview, fund performance comparison, investor analytics, and SIP/market trend visualizations.

2. Data Sources

The analysis was built on ten structured datasets, each representing a distinct aspect of the mutual fund ecosystem:

1. Fund Master (01_fund_master.csv) — Scheme-level metadata for 40 funds including AMFI code, scheme name, category (Equity/Debt), sub-category, fund house, plan type, expense ratio, and benchmark index.
2. NAV History (02_nav_history.csv) — Daily Net Asset Value records for all 40 schemes from January 2022 to May 2026, totaling approximately 46,000 data points.

3. AUM by Fund House (03_aum_by_fund_house.csv) — Quarterly Assets Under Management and scheme counts, broken down by AMC.
4. Monthly SIP Inflows (04_monthly_sip_inflows.csv) — Industry-wide SIP inflow figures, SIP AUM, and year-over-year growth rates.
5. Category Inflows (05_category_inflows.csv) — Monthly net inflows by fund category (Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS, Debt categories, etc.) for FY24-FY25.
6. Industry Folio Count (06_industry_folio_count.csv) — Monthly investor folio counts, segmented by equity, debt, hybrid, and other categories.
7. Scheme Performance (07_scheme_performance.csv) — Pre-computed return, risk, Sharpe Ratio, and rating metrics per scheme.
8. Investor Transactions (08_investor_transactions.csv) — 32,778 individual transaction records including SIP, Lumpsum, and Redemption types, with investor demographics (state, city tier, age group, income).
9. Portfolio Holdings (09_portfolio_holdings.csv) — Fund portfolio composition snapshots.
10. Benchmark Indices (10_benchmark_indices.csv) — Daily closing values for seven market indices: Nifty 50, Nifty 100, Nifty Midcap 150, BSE Smallcap, Nifty 500, CRISIL Liquid, and CRISIL Gilt.

All data was provided in CSV format and represents a synthetic but realistic simulation of Indian mutual fund industry data, following AMFI conventions for scheme naming, NAV reporting, and categorization.

3. ETL Design

The data pipeline followed a structured Extract-Transform-Load approach:

Extract: Raw CSV files were ingested representing fund master data, live and historical NAV feeds, transactional records, and market benchmarks. A separate live NAV fetch script pulled real-time data from the mfapi.in public API for six sample schemes as a proof-of-concept for live data integration.

Transform: Data cleaning steps included forward-filling missing NAV values on non-trading days, validating date ranges for consistency across tables, flagging single-day return anomalies exceeding 20% (a data-quality safeguard), and converting text-based month fields into proper date types to support time-intelligence calculations.

Load: Cleaned data was loaded into two parallel destinations:

(a) A SQLite star-schema database with fact tables (fact_nav, fact_transactions, fact_performance, fact_aum) and dimension tables (dim_fund, dim_date), enabling efficient SQL-based analytical queries.

(b) A Power BI data model with a central Date dimension table (built via DAX CALENDAR function) connected to all date-based fact tables, and a shared amfi_code key linking fund-level tables (NAV history, scheme performance, transactions, portfolio holdings) back to the fund master dimension.

This two-destination approach allowed SQL-based ad-hoc analysis (ten analytical queries covering window functions, IQR-based anomaly detection, and net flow analysis) alongside interactive Power BI dashboarding for business-facing consumption.

4. EDA Findings

Fund Universe Composition: The 40-scheme universe skews heavily toward equity (34 schemes, 85%) versus debt (6 schemes, 15%). Within equity, Large Cap is the dominant sub-category (14 schemes), followed by Mid Cap (7) and Small Cap (6). Expense ratios across the universe range from 0.55% to 1.64%, consistent with typical Indian direct/regular plan structures.

Daily Return Distribution: Pooled daily returns across all 40 schemes were examined for data quality. No single-day moves exceeding 20% were detected, indicating a clean dataset free of stock-split-style anomalies or NAV recording errors. The distribution was approximately centered near zero with the expected fat-tailed shape typical of financial return series.

Industry Trends: Industry-wide AUM data (2022–2025) shows a generally upward trajectory, consistent with the broader growth narrative of Indian mutual funds during this period. SIP inflows reached ₹31,002 crore in the most recent month, reflecting continued retail participation via systematic investing. Total investor folios stood at 26.12 crore, split across equity (largest share), debt, hybrid, and other categories.

Category Inflow Patterns: Examining FY24–FY25 category-level net inflows revealed that certain categories (typically Flexi Cap, Mid Cap, and Small Cap in the broader Indian market context) tend to attract disproportionately higher net flows relative to their scheme counts, suggesting continued retail preference for higher-growth-potential categories during this period.

Investor Demographics: Transaction records (32,778 total) spanning 12 states, 5 age groups, and 2 city tiers (T30/B30) allowed segmentation of investment behavior. SIP transactions formed a distinct behavioral pattern compared to Lumpsum and Redemption activity, with variation in average ticket size observed across age groups.

5. Performance Analysis

Composite Fund Scorecard: The weighted 0–100 scorecard (30% 3yr return, 25% Sharpe, 20% Alpha, 15% expense ratio, 10% max drawdown) ranked UTI Mid Cap Fund (88.19), HDFC Top 100 Fund (88.13), and Axis Small Cap Fund (85.25) as the top three schemes overall. Notably, several top-ranked funds carried negative or modest raw 3-year returns, reflecting the composite methodology's emphasis on risk-adjusted quality (Sharpe, Alpha, cost efficiency) rather than raw return alone.

Sharpe Ratio Rankings: Mirae Asset Large Cap Fund led on risk-adjusted return (Sharpe 1.45), followed by Kotak Flexicap Fund (1.31) and Mirae Asset Tax Saver Fund (1.23), indicating these funds delivered the most favorable return-per-unit-of-volatility over the analysis window.

Alpha and Beta vs Nifty 100: SBI Small Cap Fund (30.3% annualized alpha), DSP Small Cap Fund (30.1%), and ICICI Pru Midcap Fund (29.3%) showed the strongest outperformance relative to the Nifty 100 benchmark. However, R-squared values across nearly all 40 funds were very low (well under 0.01 in most cases), indicating minimal statistical correlation between fund returns and the Nifty 100 benchmark in this dataset — a finding that warrants further validation against the source data, as real-world equity funds typically show much higher benchmark correlation (R-squared often 0.7+).

Maximum Drawdown: SBI Small Cap Fund (Direct Plan) recorded the deepest drawdown at -52.6%, with a prolonged decline from January 2023 to October 2025. Axis Small Cap Fund and ABSL Small Cap Fund followed with drawdowns of -51.7% and -35.4% respectively. Small Cap category funds were disproportionately represented among the worst drawdowns, consistent with their higher-volatility risk profile.

Tracking Error: Among the top 5 scorecard funds, tracking error against Nifty 100 ranged from 18.6% (Axis Bluechip Fund) to 29.3% (ABSL Small Cap Fund) on an annualized basis — relatively high figures that, combined with the low R-squared findings above, suggest these funds' return patterns diverge substantially from broad-market index movements in this dataset.

6. Dashboard Screenshots

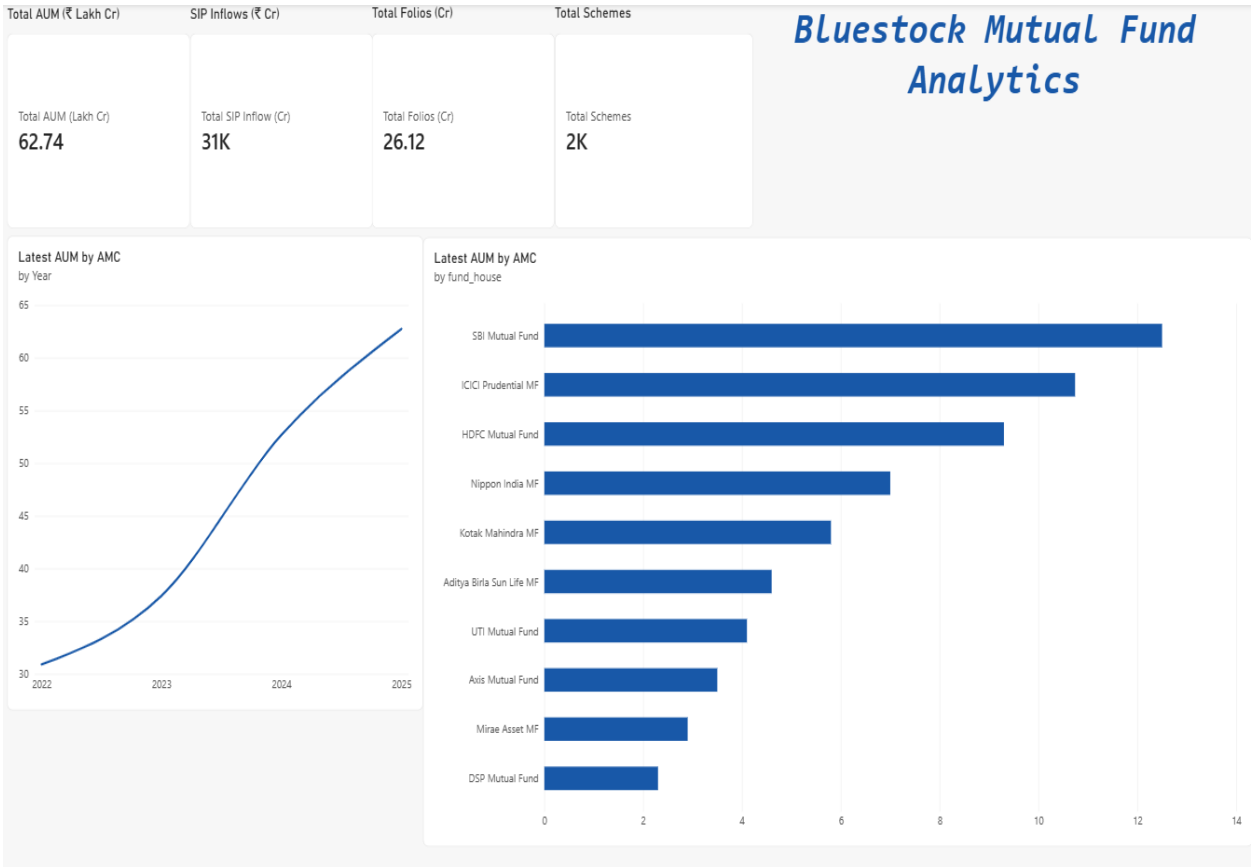


Figure 1: Industry Overview Dashboard

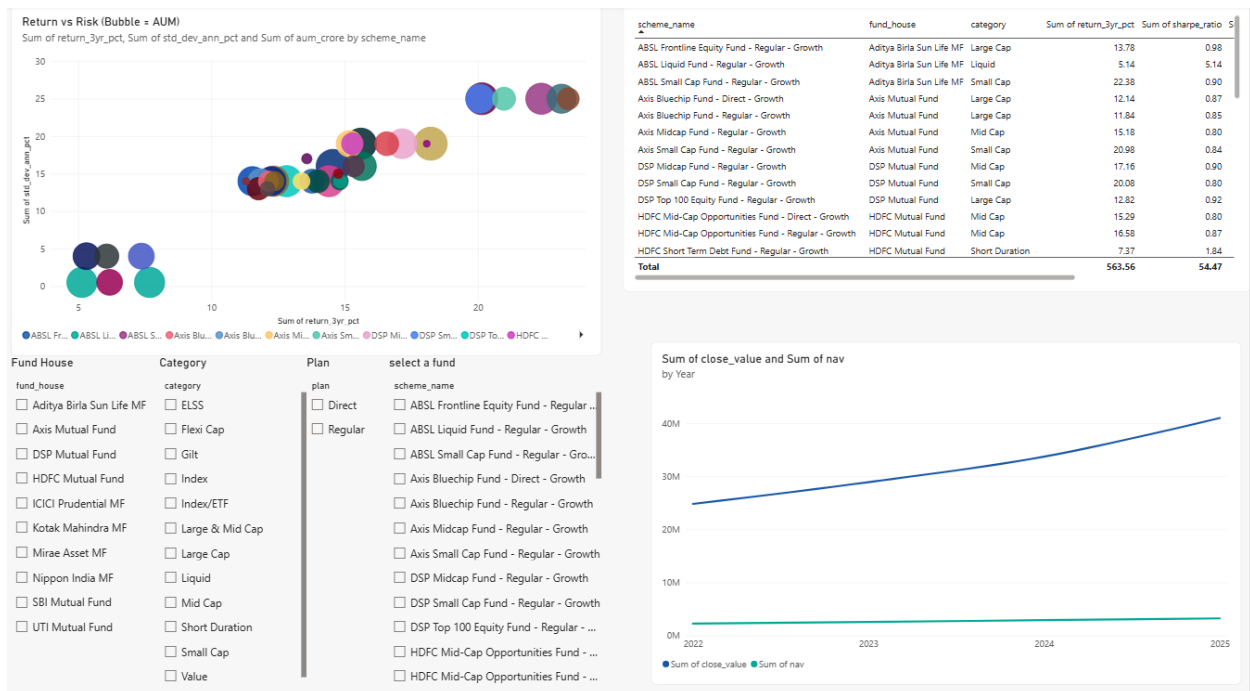


Figure 2: Page2_FundPerformance.png

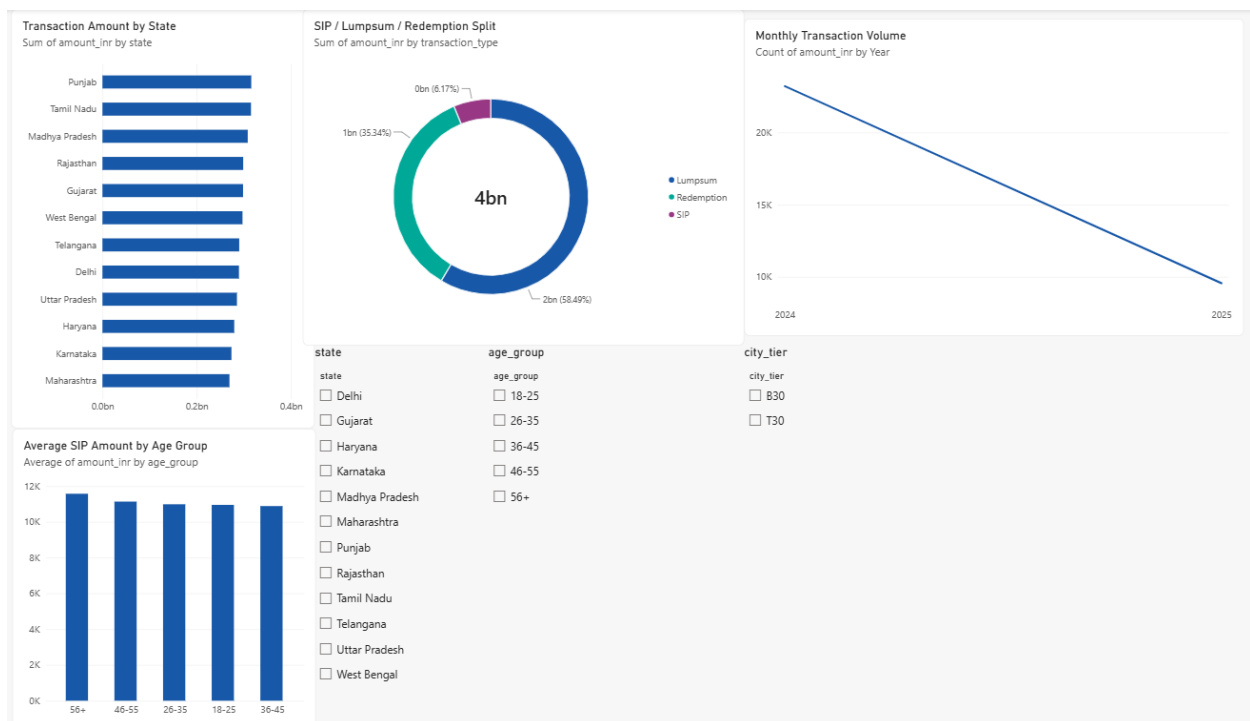


Figure 3: Page3_InvestorAnalytics.png

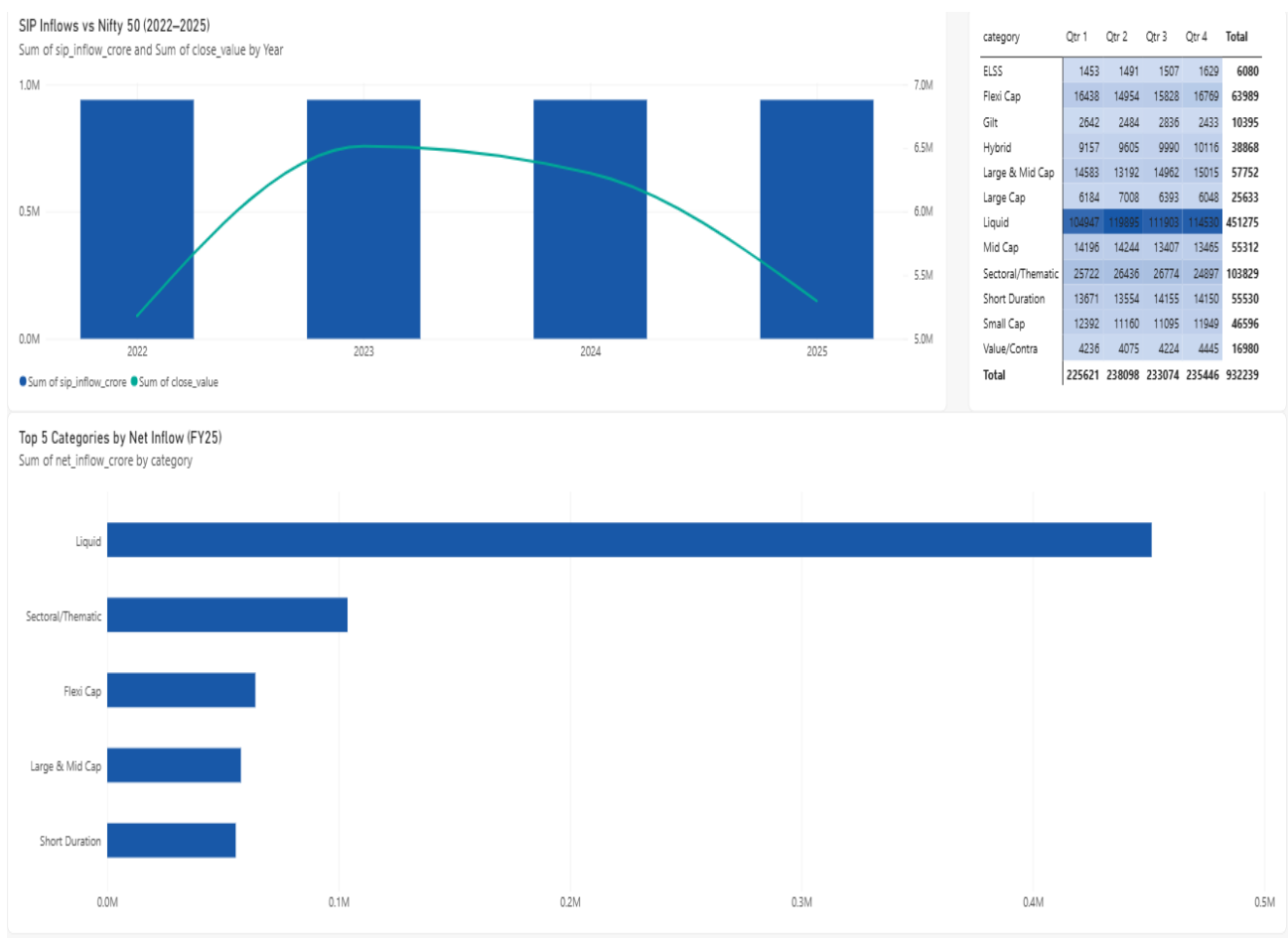


Figure 4: Page4_SIPMarketTrends.png

7. Limitations

Several limitations should be considered when interpreting this analysis:

Data Nature: The dataset used for this project, while structurally realistic and AMFI-consistent, is synthetic. Absolute figures (returns, AUM, alpha/beta values) should be treated as illustrative of methodology rather than as live market intelligence.

Low Benchmark Correlation: As noted in Section 5, R-squared values between fund returns and the Nifty 100 benchmark were unusually low across nearly all 40 schemes, including large-cap equity funds that would typically show strong benchmark tracking. This suggests either a data generation characteristic of the synthetic dataset or a data alignment issue that would benefit from further investigation before drawing real-world conclusions from alpha/beta figures.

Static Risk-Free Rate: The Sharpe and Sortino Ratio calculations used a fixed 6.5% annual risk-free rate (RBI repo rate proxy) for the entire analysis period. In reality, the repo rate fluctuated during 2022–2026, and a time-varying risk-free rate would produce more precise risk-adjusted metrics.

Survivorship and Scheme Coverage: The 40-scheme universe, while diverse across categories, represents a small subset of the 1,500+ schemes in the actual Indian mutual fund industry (per the AUM-by-fund-house data). Findings should not be generalized to the full industry.

Category Imbalance: The universe skews heavily toward equity (85%) versus debt (15%) funds, which may bias aggregate findings toward equity-market dynamics.

8. Recommendations

Based on the findings of this analysis, the following recommendations are proposed:

1. **Validate Benchmark Data Alignment:** Before using the alpha/beta figures for any investment decision-making, investigate the low R-squared values found across the fund universe — check for date misalignment, incorrect benchmark mapping per fund, or data generation artifacts in the source NAV/benchmark series.
2. **Incorporate Time-Varying Risk-Free Rate:** Future iterations of this analysis should pull historical RBI repo rate data month-by-month rather than using a single static rate, improving the precision of Sharpe and Sortino calculations.
3. **Expand the Fund Universe:** Extending analysis to a larger, more representative sample of schemes (or the full industry dataset) would improve the generalizability of category-level and demographic findings.
4. **Automate Data Refresh:** Given the live NAV fetch capability already built in Day 1 of this project (via the mfapi.in API), a scheduled refresh pipeline could keep the SQLite database and Power BI dashboard current without manual re-import.

5. Add Debt-Fund-Specific Benchmarking: Since 6 of the 40 schemes are Debt category funds, future work should benchmark them against CRISIL Gilt or CRISIL Liquid indices (already present in the dataset) rather than Nifty 100, for a more meaningful alpha/beta comparison.

6. Publish and Share the Dashboard: Publishing the Power BI dashboard to Power BI Service (free tier) would allow stakeholders to access live, interactive views without needing Power BI Desktop installed locally.

Appendix A: Methodology & Formulas

Daily Return: $\text{daily_return} = (\text{NAV}_t / \text{NAV}_{t-1}) - 1$

CAGR: $\text{CAGR} = (\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(1/n)} - 1$, where n is the holding period in years, calculated using the actual calendar-day span between the start and end dates.

Sharpe Ratio: $\text{Sharpe} = (\text{Rp} - \text{Rf}) / \sigma(\text{Rp}) \times \sqrt{252}$, where Rp is the mean daily return, Rf is the daily risk-free rate (6.5% annual, RBI repo rate proxy), $\sigma(\text{Rp})$ is the standard deviation of daily returns, and $\sqrt{252}$ annualizes the ratio based on trading days per year.

Sortino Ratio: Same formula as Sharpe, but the denominator uses only the standard deviation of negative-return days (downside deviation), penalizing only harmful volatility rather than all volatility.

Alpha and Beta: Estimated via Ordinary Least Squares (OLS) linear regression of each fund's daily returns against the Nifty 100 benchmark's daily returns, using `scipy.stats.linregress`. Beta is the regression slope (sensitivity to benchmark movement); Alpha is the regression intercept, annualized by multiplying by 252 (representing return unexplained by benchmark movement).

Maximum Drawdown: For each point in time, $\text{drawdown} = (\text{NAV}_t / \text{running_maximum_NAV}) - 1$. Maximum drawdown is the most negative value of this series across the fund's full history, with peak and trough dates identified as the last all-time-high before, and the lowest point during, the decline.

Fund Scorecard: A composite 0–100 score computed as: $30\% \times \text{percentile_rank}(\text{3yr CAGR}) + 25\% \times \text{percentile_rank}(\text{Sharpe Ratio}) + 20\% \times \text{percentile_rank}(\text{Alpha}) + 15\% \times \text{percentile_rank}(\text{inverse Expense Ratio}) + 10\% \times \text{percentile_rank}(\text{inverse Max Drawdown})$. Percentile ranks are calculated across all 40 schemes, with expense ratio and max drawdown inverted so that lower cost and shallower drawdown score higher.

Tracking Error: Calculated as the annualized standard deviation of the daily difference between fund returns and benchmark returns: $\text{tracking_error} = \sigma(R_{\text{fund}} - R_{\text{benchmark}}) \times \sqrt{252}$.

Appendix B: Complete Fund Scorecard (All 40 Schemes)

	Overall Rank	Category	Fund Score	CAGR 3yr	Sharpe	Alpha	Expense Ratio	Max Draw down
UTI Mid Cap Fund - Regular - Growth	1	Equity	88.1875	-0.00767	-0.20583	0.028969	1.51	-0.28001
HDFC Top 100 Fund - Regular Plan - Growth	2	Equity	88.125	0.012924	-0.20152	0.037476	1.55	-0.24734
Axis Small Cap Fund - Regular - Growth	3	Equity	85.25	-0.11703	-0.07597	0.048016	1.38	-0.51678
ABSL Small Cap Fund - Regular - Growth	4	Equity	80.4375	-0.04151	0.162661	0.108971	1.53	-0.35447
Axis Bluechip Fund - Regular - Growth	5	Equity	79.9375	0.005258	0.030785	0.068995	1.64	-0.14402
SBI Small Cap Fund - Direct Plan - Growth	6	Equity	78.5625	-0.01337	-0.05719	0.048824	0.72	-0.52574
Kotak Emerging Equity Fund - Regular - Growth	7	Equity	74.875	0.077542	0.076	0.078044	1.56	-0.24004
HDFC Short Term Debt Fund - Regular - Growth	8	Debt	70.625	0.039155	-0.56709	0.042818	0.56	-0.04308
ABSL Liquid Fund - Regular - Growth	9	Debt	69.625	0.063143	0.81557	0.060861	0.79	-0.00162
SBI Magnum Gilt Fund - Regular Plan - Growth	10	Debt	68.75	0.05839	0.22658	-0.056209	0.77	-0.04329

Nippon India
Gilt Securities
Fund -

Regular -			0.04061	-	0.05074		
Growth	11	Debt	67.875	3	0.35663	8	0.55 -0.08316

Nippon India
Small Cap
Fund -

Regular -			64.187		0.44843	0.17500	
Growth	12	Equity	5	0.08006	4	7	1.53 -0.23345

Kotak
Bluechip Fund
- Regular -

Growth	13	Equity	63.125	0.17263	0.49427	0.13042	1.59 -0.17574
					3	9	

Kotak Liquid
Fund -
Regular -

Growth	14	Debt	61.375	0.06694	-	0.06455	
				4	0.08875	7	0.6 -0.00116

HDFC Mid-
Cap
Opportunities
Fund - Direct -

Growth	15	Equity	60.375	0.11242	0.30061	0.11989	
				5	2	1	0.78 -0.21117

DSP Top 100
Equity Fund -
Regular -

Growth	16	Equity	58.812	0.18039	0.46403	0.13173	
			5	7	7	2	1.54 -0.14845

UTI Flexi Cap
Fund -
Regular -

Growth	17	Equity	56.312	0.25549	0.61951	0.16211	
			5	7	8	3	1.64 -0.2154

Axis Bluechip
Fund - Direct -
Growth

	18	Equity	54.875	0.20811	0.12961	0.08232	
				6	4	8	0.75 -0.21751

ICICI Pru
Liquid Fund -
Regular -

Growth	19	Debt	54.5	0.07392	0.49572	0.06746	
				5	3	2	0.74 -0.00098

ICICI Pru
Value
Discovery
Fund -

Regular -				0.18417	0.64887	0.16253	
Growth	20	Equity	51.375	4	9	9	1.41 -0.19826

Mirae Asset
Emerging
Bluechip Fund

	21	Equity	49.062	0.14744	0.92628		
			5	4	7	0.20873	1.52 -0.15273

- Regular -
Growth
ICICI Pru
Bluechip Fund

- Regular -

Growth
UTI Nifty 50
Index Fund -

Regular -

Growth

Nippon India

Large Cap

Fund - Direct -

Growth

Nippon India

ETF Nifty 50

BeES

HDFC Top 100

Fund - Direct

Plan - Growth

SBI Bluechip

Fund - Direct

Plan - Growth

Nippon India

Large Cap

Fund -

Regular -

Growth

DSP Small Cap

Fund -

Regular -

Growth

DSP Midcap

Fund -

Regular -

Growth

SBI Small Cap

Fund -

Regular Plan -

Growth

ABSL

Frontline

Equity Fund -

Regular -

Growth

Mirae Asset

Tax Saver

Fund -

22	Equity	48.875	0.13975	0.79797 3	0.17703 3	1.42	-0.13898
23	Equity	47.5	0.19662 4	0.81709 9	0.17048 8	1.57	-0.1086
24	Equity	43.937 5	0.21084 7	0.64520 7	0.15643 8	0.72	-0.1863
25	Equity	43.5	0.20003 2	0.66485 7	0.15136 4	0.89	-0.11651
26	Equity	43.25	0.18429 9	0.77579	0.17729 8	0.92	-0.152
27	Equity	36.5	0.16263 4	0.95327 9	0.19868 6	0.66	-0.11803
28	Equity	35.937 5	0.22646 6	1.08165 9	0.21829 4	1.51	-0.17414
29	Equity	35.812 5	0.26993 5	0.94979 6	0.30057 9	1.52	-0.31172
30	Equity	35.75	0.26863 1	1.13212 2	0.26598 6	1.61	-0.17248
31	Equity	35.125	0.26663 1	0.94530 8	0.30337	1.43	-0.28706
32	Equity	34.312 5	0.28960 2	1.02721 3	0.21399 8	1.6	-0.11292
33	Equity	28.812 5	0.29171 4	1.23493	0.28270 4	1.6	-0.16397

Regular - Growth SBI Bluechip Fund - Regular Plan - Growth	34	Equity	27.687 5	0.30448 6	1.20826 7	0.23201	1.54	-0.15012
Axis Midcap Fund - Regular - Growth	35	Equity	25.5	0.35102 5	0.99823 1	0.26076 7	1.38	-0.20961
ICICI Pru Bluechip Fund - Direct - Growth	36	Equity	22.5	0.32478 9	1.02652 4	0.21194 8	0.8	-0.12588
HDFC Mid- Cap Opportunities Fund - Regular - Growth	37	Equity	21.75	0.32434	1.09369 9	0.27195 4	1.38	-0.16217
Kotak Flexicap Fund - Regular - Growth	38	Equity	20.5	0.29575 1	1.30674 4	0.27330 5	1.45	-0.12974
ICICI Pru Midcap Fund - Regular - Growth	39	Equity	20.25	0.31769 2	1.18010 1	0.29263 6	1.36	-0.18189
Mirae Asset Large Cap Fund - Regular - Growth	40	Equity	16.25	0.33992	1.44829 1	0.26983 8	1.46	-0.11266